

ReneWind Project

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Contents/Agenda

- Executive Summary
- Business Problem Overview and Solution Approach
- Data Overview
- EDA Results
- Data Preprocessing
- Model Building
- Model Performance Summary
- Test Performance of Best Model
- Appendix

Executive Summary - Insights

- About 5% of turbines are failing and are very costly as a result
- ReneWind needs a model that can catch these failures in order to save money and not send out crews for 'false alarms'
- There are 18 missing values in V1 and V2 and therefore making the data set incomplete

Executive Summary - Recommendations

- We need to create a model that focuses on recall to catch the 5% failures and therefore saving the company money
- By creating a model with a focus on recall we can ensure it will catch most of all failures and have repair crews on site to repair them instead of having to replace the turbines
- To prevent data leakage we will impute missing values after we split the train data into train and validation sets. We will impute the missing values using the median

Business Problem

Overview and Solution Approach

ReneWind is a company working on improving the machinery/processes involved in the production of wind energy. They have collected data from generator failure of wind turbines using sensors. We need to build various classification models and find the best model that will help identify failures. This will help ReneWind repair the turbines before they fail or break thus reducing overall cost.

EDA Results

Data Overview

- We see that the training set has 20,000 rows and 41 columns
- The test set has 5,000 rows and 41 columns
- There are no duplicate values in our datasets
- In the training data there are 18 missing values in V1 and 18 missing in V2

EDA Results

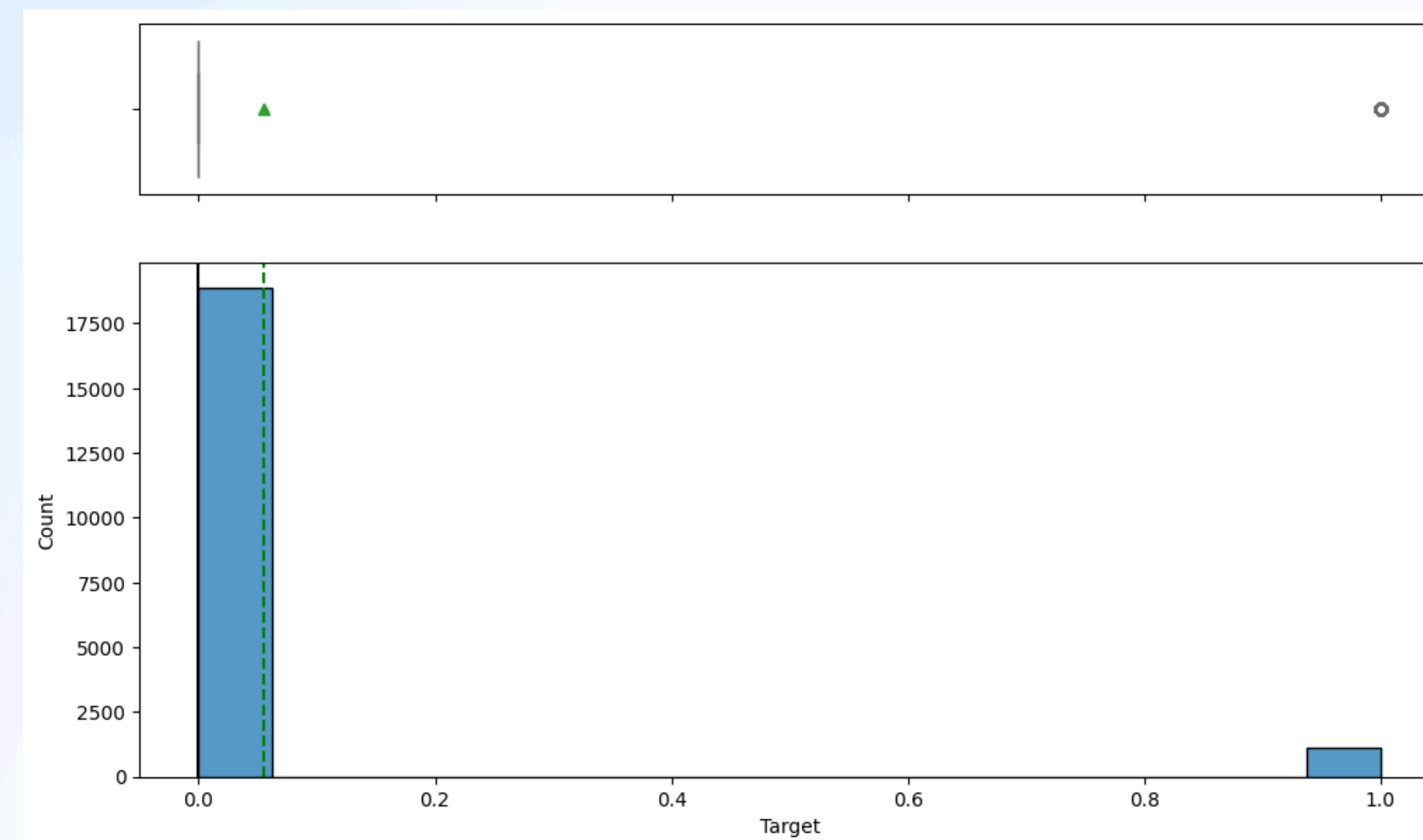
Univariate Analysis V1-V40

- Most of the variables are symmetrical. There are some that are right and left skewed which I will list below
- We see that V1, V3, V35, V37 are right skewed and V8, V16, V17 are left skewed
- The other variables are all symmetrical

EDA Results

Target Variable

About 18,000 of the target variables below have 'no failure'. About 2,000 do have failure. 94% of the target variables in the train and testing data have 'no failure'.



EDA Results

Bivariate Analysis

- V29 & V11(0.81), V24 & V32(0.83), V21 & V16(0.84), V15 & V7(0.87) all have some of the highest positive correlation
- V14 & V2 (-0.85), V23 & V3 (-0.79), V27 & V32(-0.77) have some of the highest negative correlation

Data Preprocessing

Data Preparation for Modeling

When we prepare the data for modeling we have to split our data into training and validation sets. The shape of the X Train data has 16,000 rows and 40 columns. The shape of the X validation data has 4,000 rows and 40 columns. Once we divide the test data into X test and Y test we can see the X test shape has 5,000 rows and 40 columns.

Data Preprocessing

Missing Value Imputation

We did have some missing values in V1 and V2 and we will impute them using the median to get rid of these missing values.

Model Building

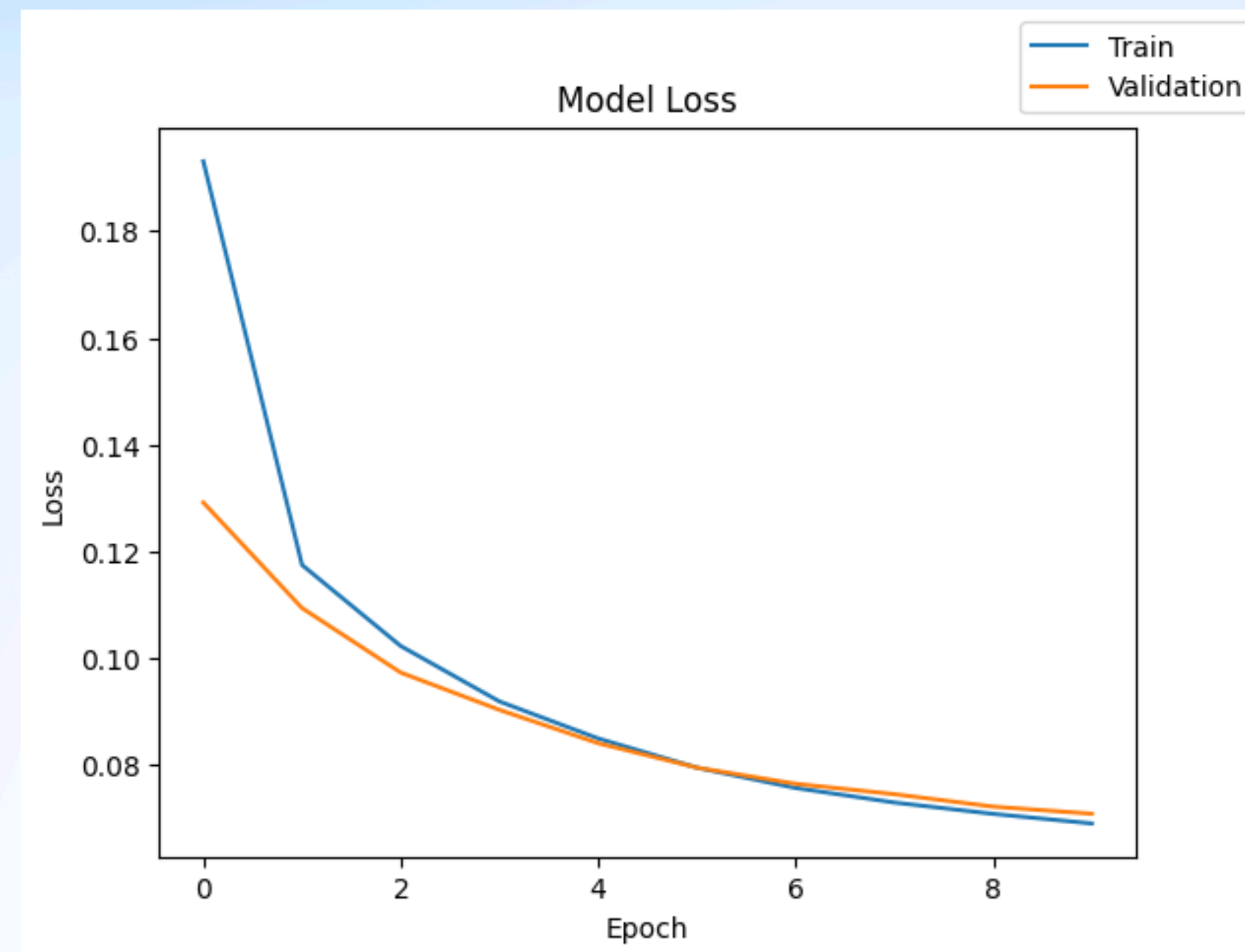
Metric of Choice - Recall

The metric we will choose for our module is Recall. The reason being is because a high recall ensures the model catches as many failures as possible. This will achieve the objective the company has which is saving money by catching the failures.

Model Building

Neural Network Model with SGD

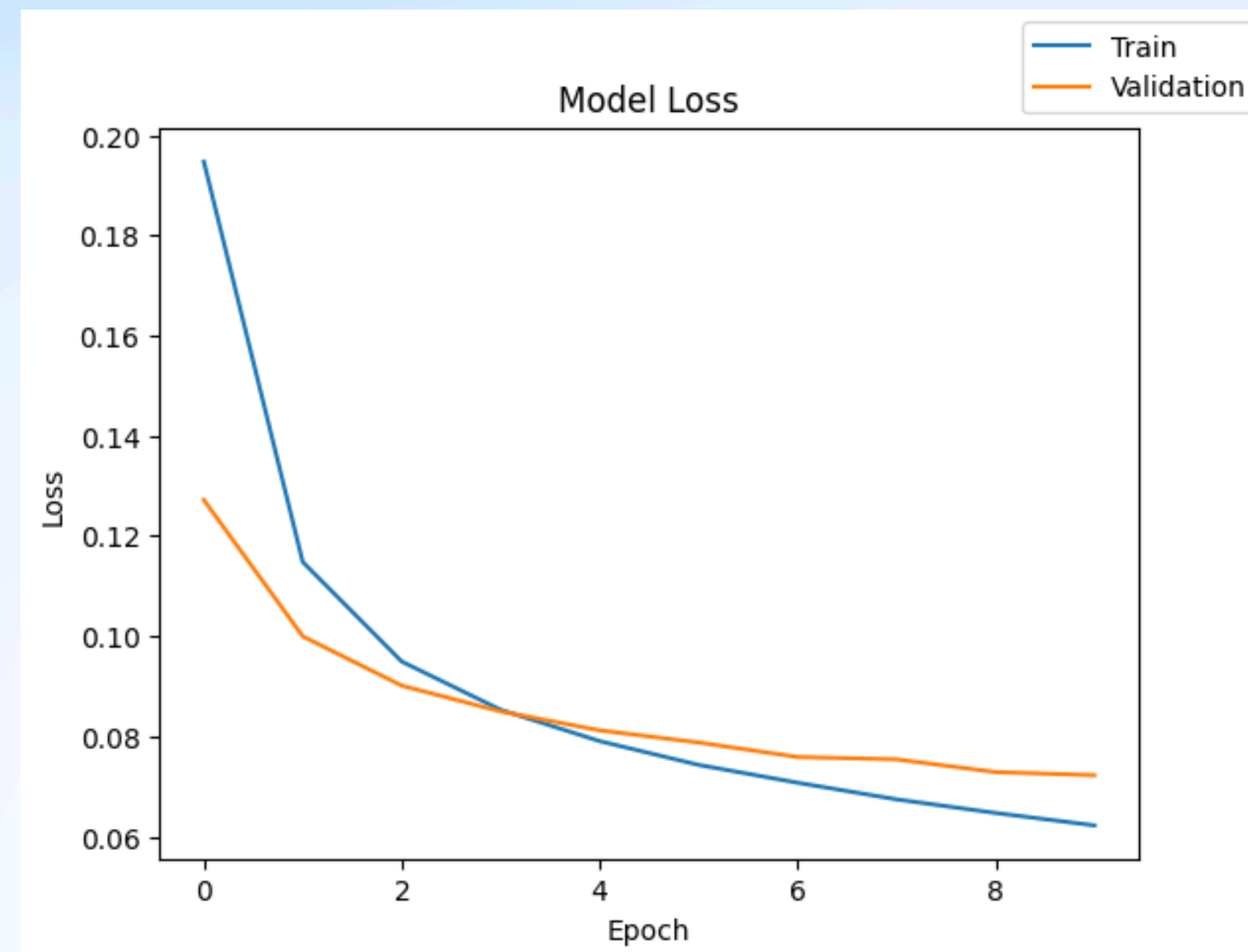
We can see below our model is performing well. As the epochs increase our model improves and is performing well on unseen data and data it was trained on. Our recall score here is .77 which means the model catches failures 77% of the time. The model is correct 77% of the time at catching failures.



Model Performance Summary

Model 1 With An Additional Hidden Layer

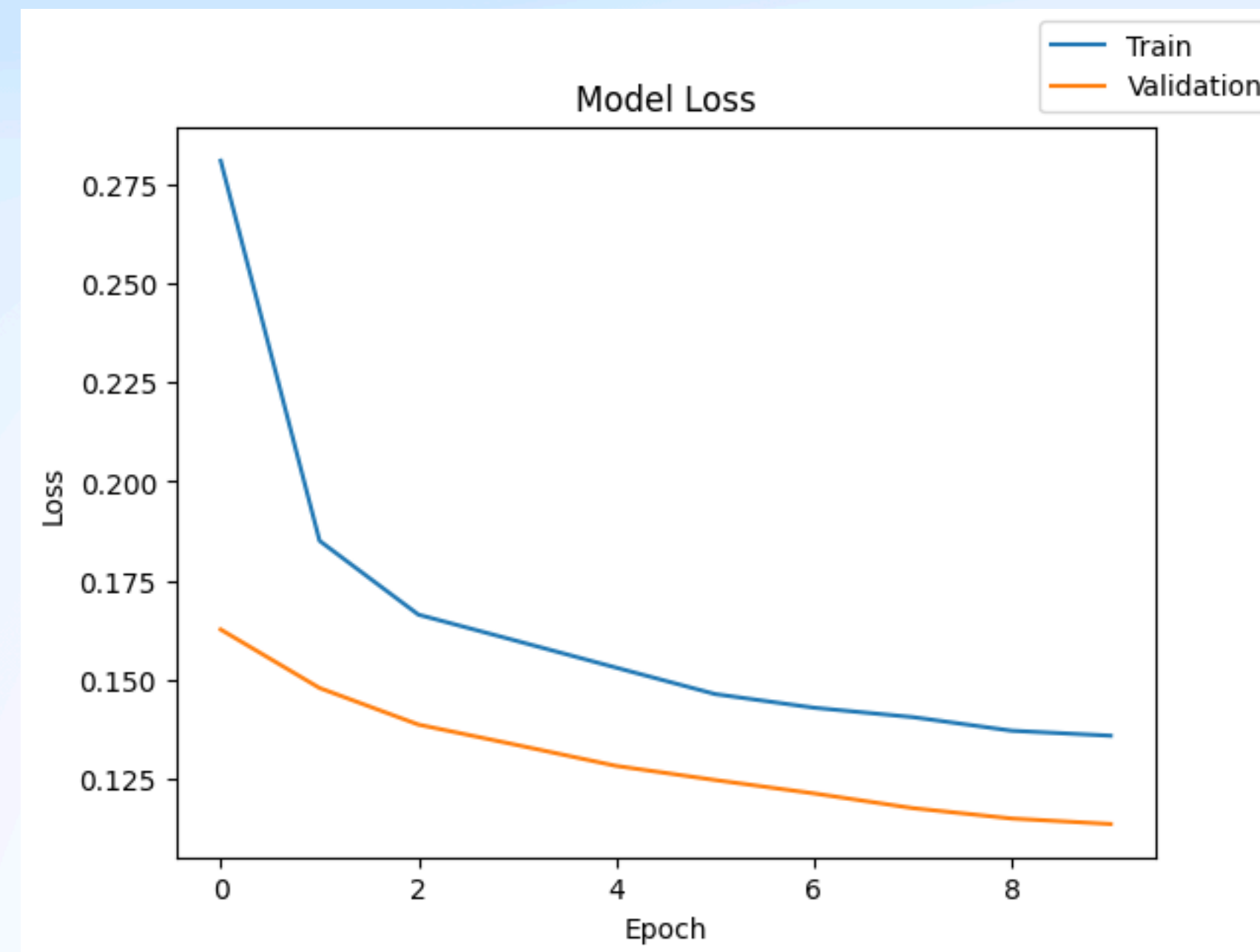
We can see below our validation set is not performing as good as our training set. Our validation set recall score is .77 which means the model is capturing 77% of all failures. Our model is a little overfitted and we need to continue to improve our model and increase the recall score.



Model Performance Summary

Model 2 With Dropout

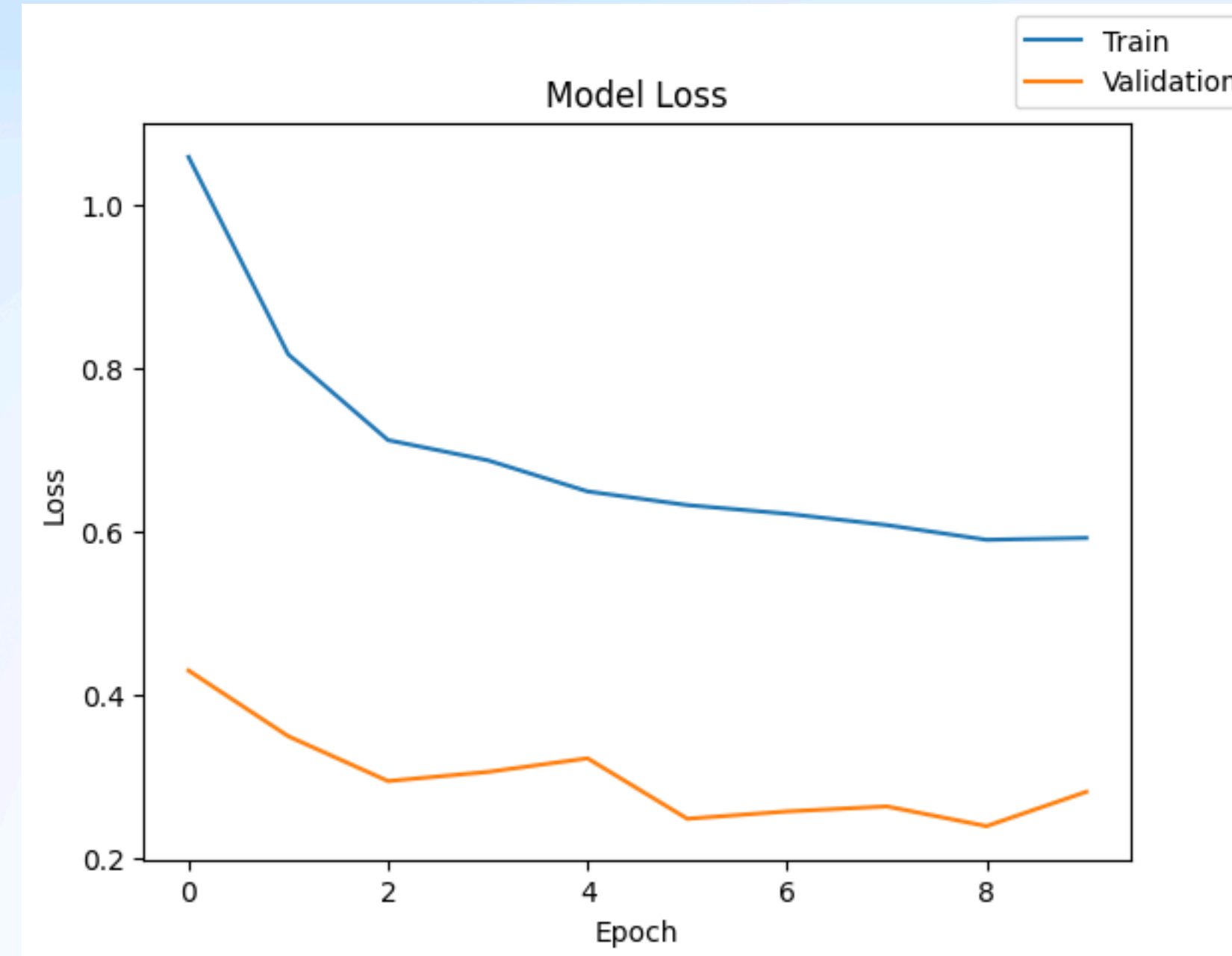
There is a huge difference in the recall score in this model compared to model 1. Our validation set recall score is 0.31 which means the model is catching 31% of all failures. We will not use this model because of the low recall score and imbalance in class distribution.



Model Performance Summary

Model 3 With Class Weights

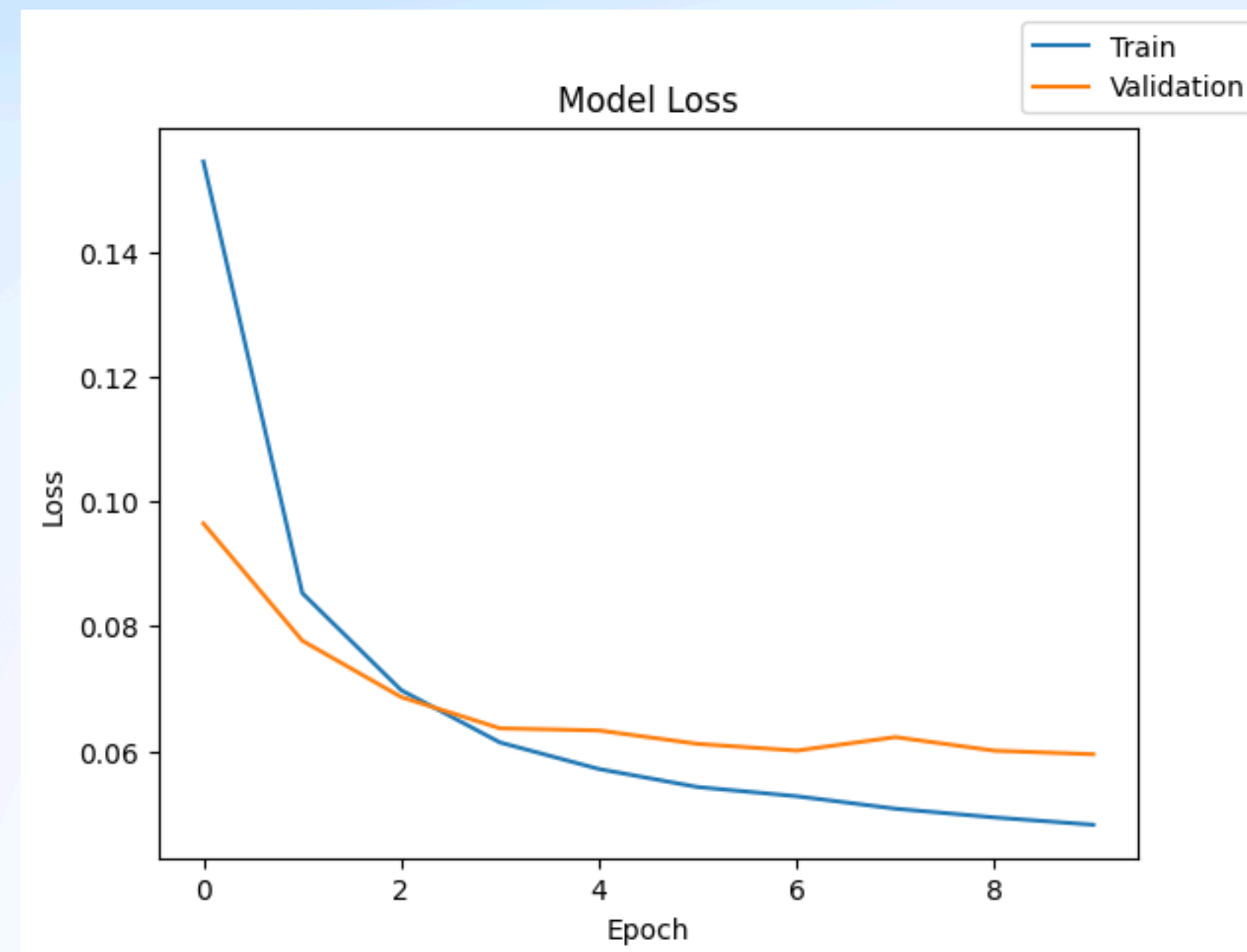
We see our model below has class weights and we were able to bring the recall up to .87 meaning our model catches 87% of all failures. This is a large improvement compared to our previous model.



Model Performance Summary

Model 4 With Adam Optimizer

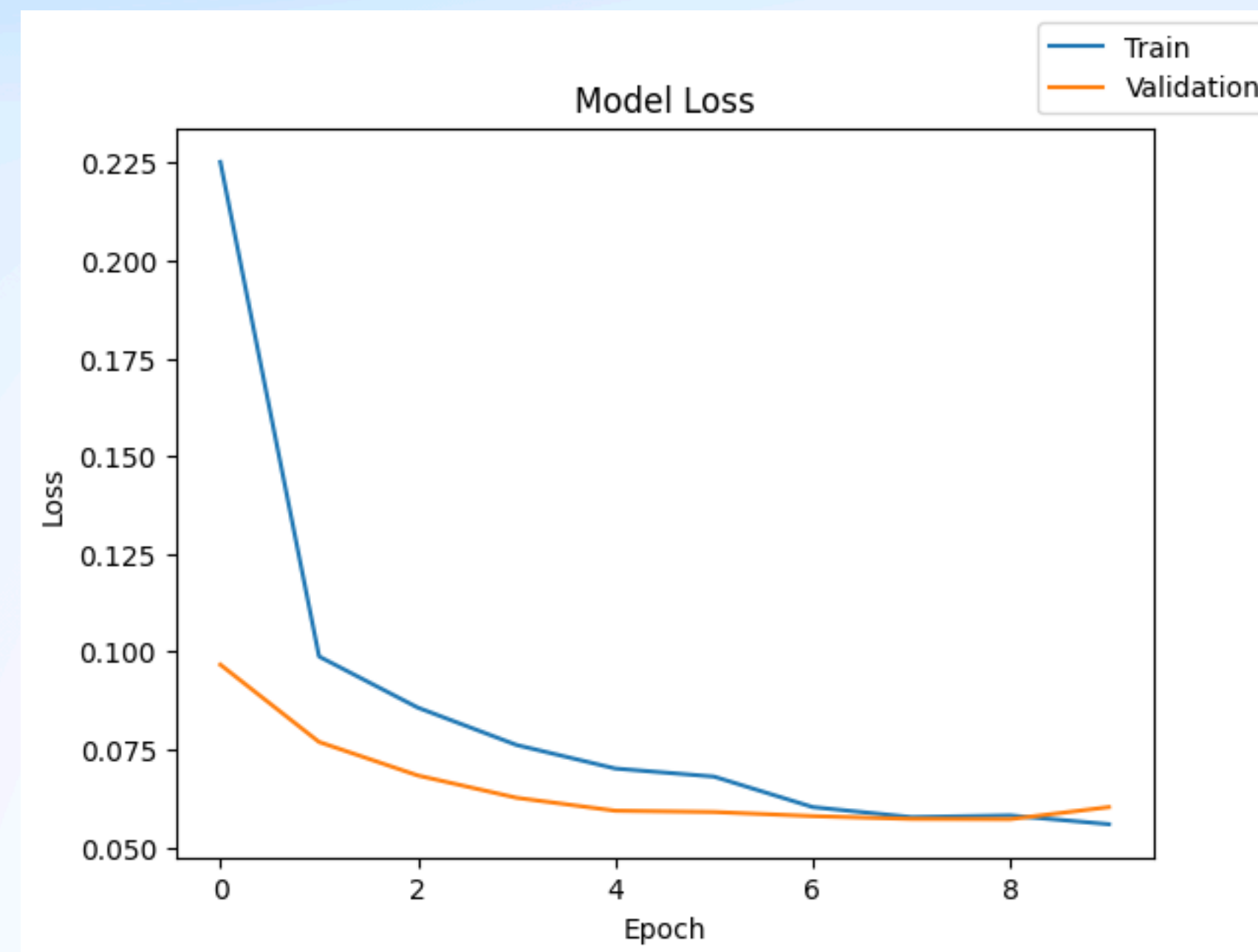
We see our model below was created using the Adam optimizer and the validation set recall score was 0.84 meaning our model catches 84% of all failures. Our model is overfitted because the validation set is performing worse than the training set.



Model Performance Summary

Model 5 With Hidden Layers, Dropout & Adam Optimizer

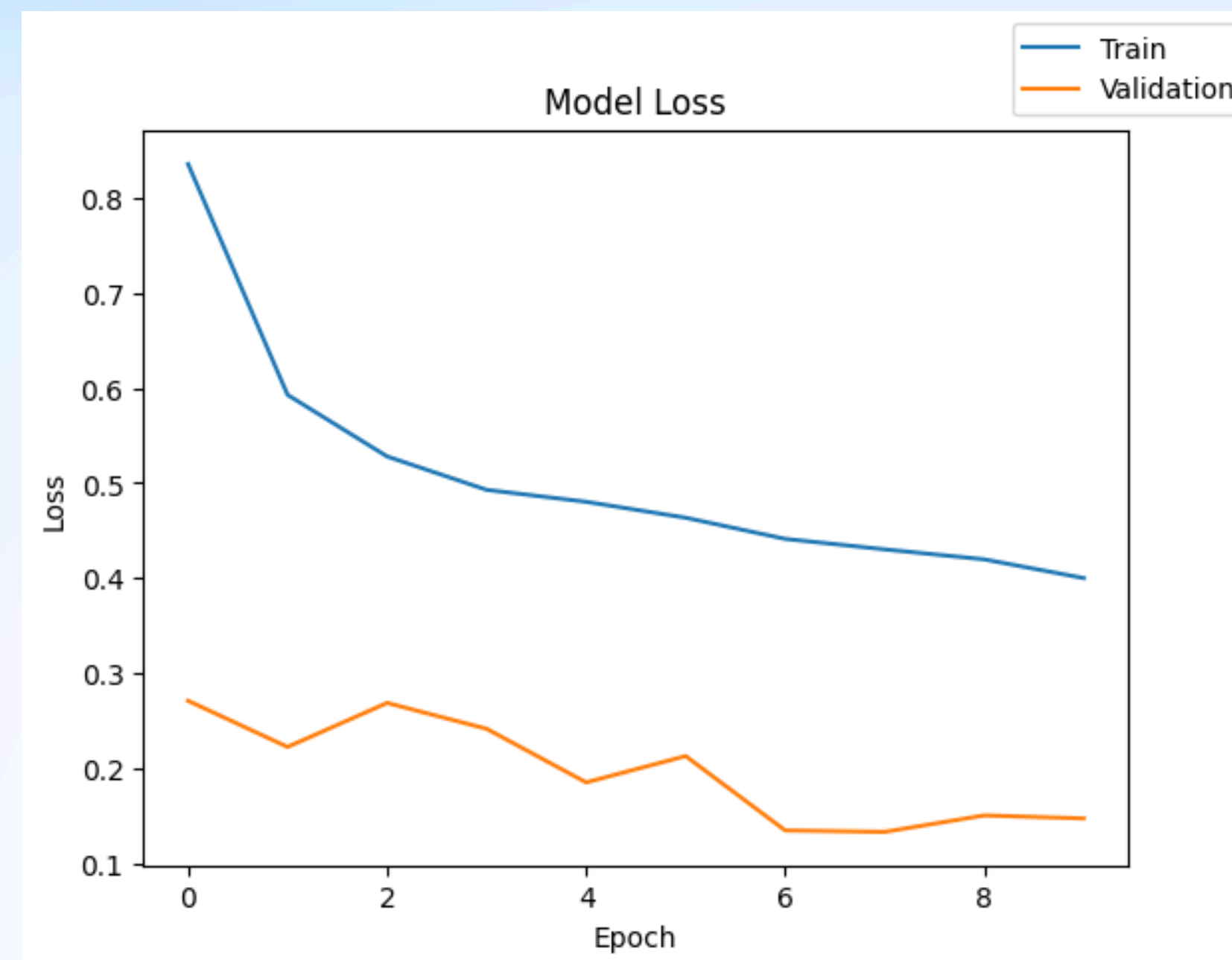
We see our model below was created using the Adam optimizer, more hidden layers and dropout at 30%. The validation set recall score was 0.82 meaning our model catches 82% of all failures. We can see our validation set is outperforming our training set up until the 8th epoch.



Model Performance Summary

Model 6 With SGD Optimizer, Dropout and Class Weights

We see our model below was using the SGD optimizer, dropout at 30% and higher weight importance to the minority class. The validation set recall score was 0.89 meaning our model catches 89% of all failures. The model is our favorite since it has the highest recall score and will save the most money for the company which is their goal.



Final Model Selection

We will look at the recall score for every model:

- Model 1: 0.77
- Model 2: 0.31
- Model 3: 0.87
- Model 4: 0.84
- Model 5: 0.82
- Model 6: 0.89

We can see that model 6 has the highest recall score. We will pick this model because a higher recall means the model is catching all true failures and in this case 89% of all failures. This will save the company a lot of money. I chose recall also because I would rather have the company repair as many failed turbines as possible and save money rather than replace them and have their spending increase.

Appendix

Bivariate analysis correction
check from slide 9:

